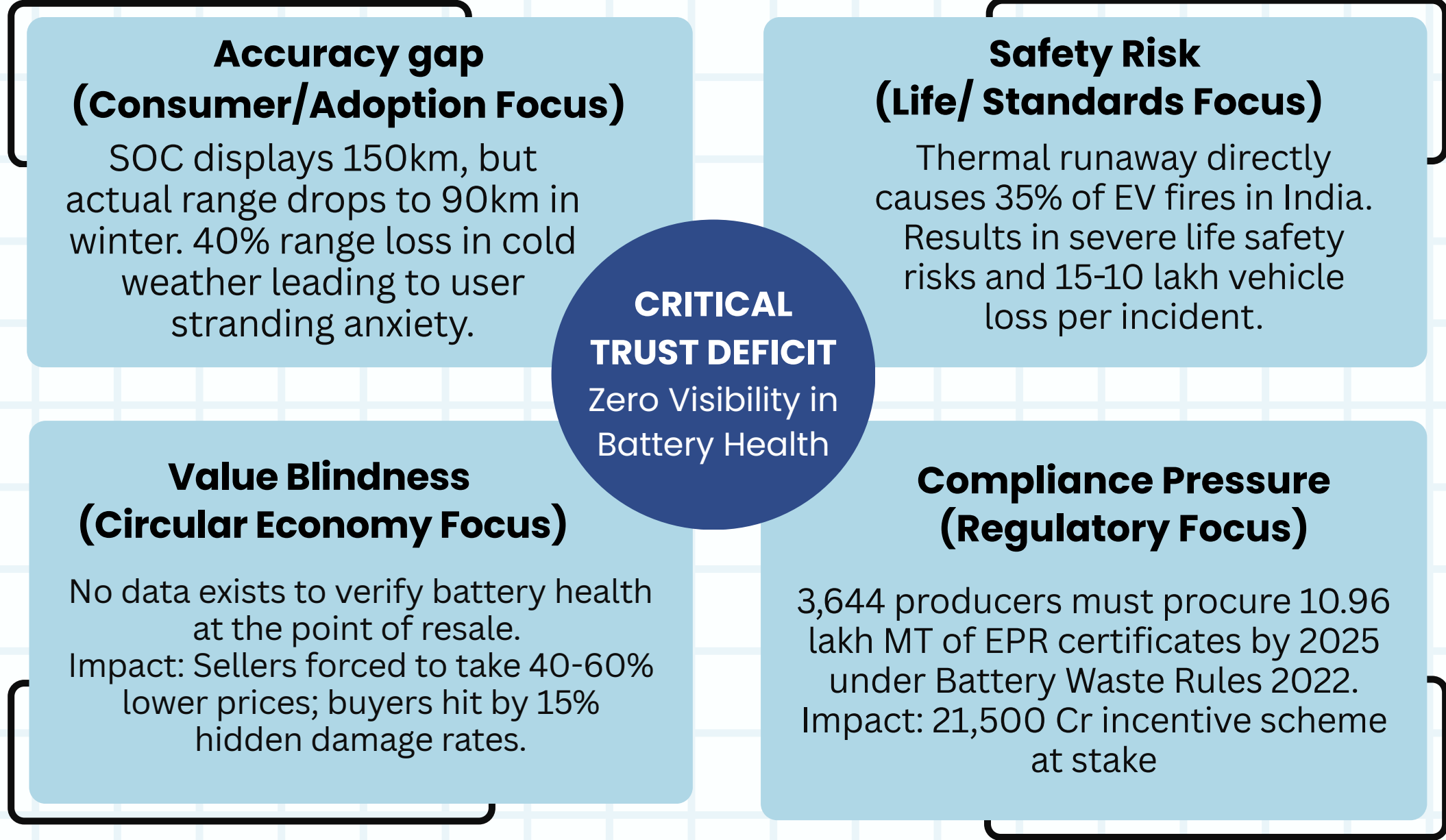




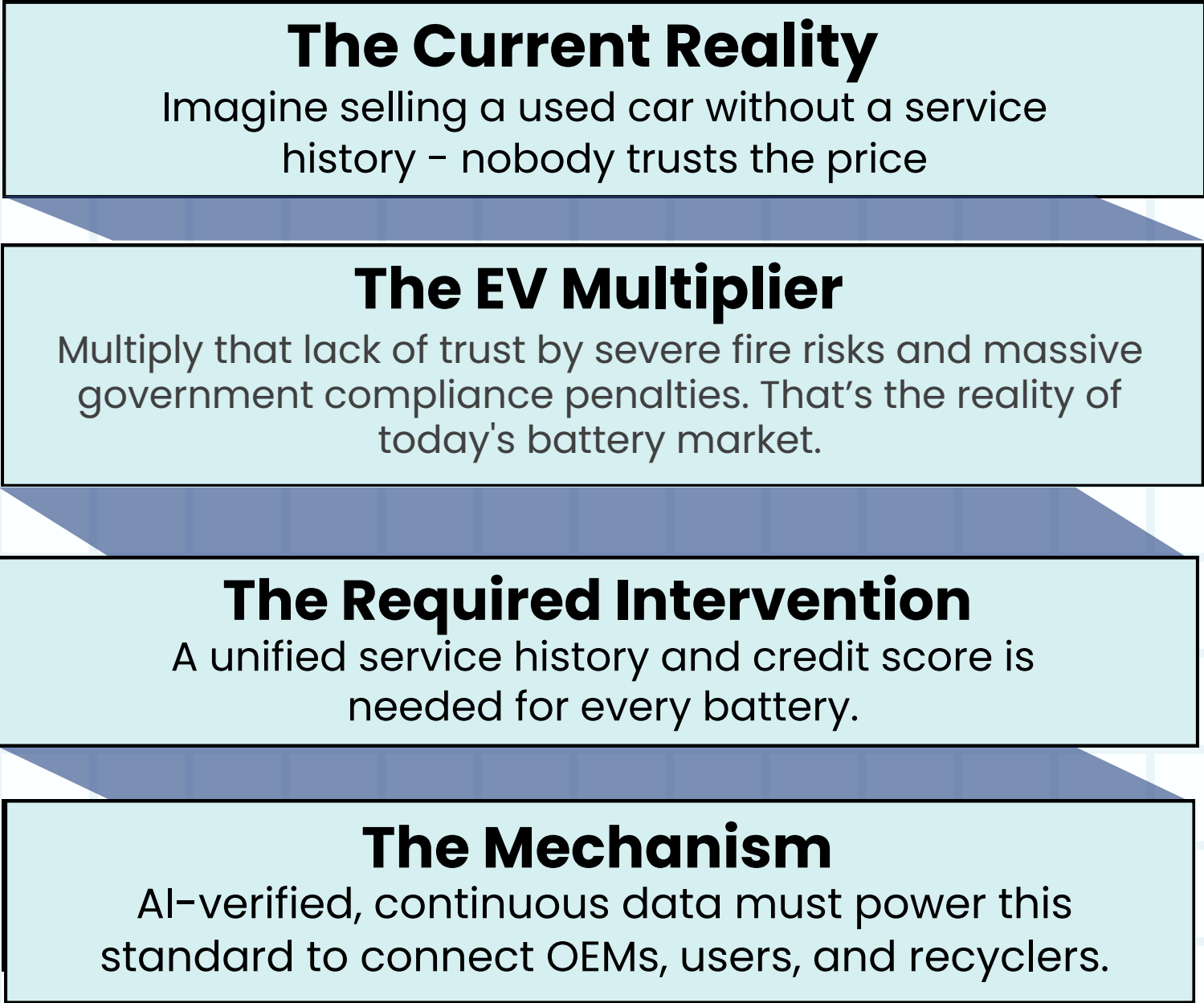
ReCell
INTELLIGENT BATTERY MANAGEMENT SYSTEM

1 Problem Identification

Fourfold EV Battery Crisis: A Systemic Failure of Data



Trust Deficit in Second-Life Battery Trading



ROOT CAUSES

No unified Data Standard

No Verified History

Multi-Parameter Ignorance

No Compliance Bridge

Proposed Solution

AI-Powered Battery Intelligence

- Accurate SOC and range prediction
- Battery health (SOH) monitoring
- Remaining useful life (RUL) forecasting
- Anomaly detection and degradation tracking

Smart Battery Optimization Engine

- Smart charging recommendations
- Temperature-aware charging optimization
- Predictive maintenance alerts
- Driving and charging behavior insights

Certified Second-Life Battery Marketplace

- Verified battery listings from users, fleets, and recyclers
- Battery condition filters (SOH, cycles, chemistry, RUL)
- Price discovery based on battery health

AI-Based Battery Certification System

- AI-powered diagnostics using BMS data
- Standardized battery health reports
- Tamper-resistant battery performance history
- Digital battery health certificates

Target Customers



URBAN EV COMMUTER

Name: Rohit Sharma

Age: 29

Location: Bangalore

Vehicle: Electric 2-wheeler / compact EV

Rohit is a young urban professional who uses an EV for daily commuting and savings. He is tech-savvy but relies entirely on the dashboard for battery information, with little visibility into real battery health or range accuracy.

THINKS

- "Is my battery degrading faster than normal?"
- "Can I trust the range shown?"
- "Will fast charging damage my battery?"

SAYS

- "The range drops too fast sometimes."
- "Why does the percentage fall suddenly?"
- "I wish I knew how healthy my battery is."

FEELS

- Anxious about unexpected breakdowns
- Confused about battery health
- Concerned about replacement cost

DOES

- Charges overnight at home
- Uses fast chargers
- Checks battery % frequently
- Searches online for advice



EV FLEET OPERATOR

Name: Priya Mehta

Age: 38

Business: 150 EV delivery vehicles

Location: Delhi NCR

Priya manages a last-mile delivery fleet where vehicles run 10–12 hours daily. Battery performance directly affects operations and profitability. She focuses on efficiency, cost control, and minimizing downtime through better maintenance and charging strategies.

THINKS

- "Which vehicles are degrading faster?"
- "When should I replace batteries before failure?"
- "Are we damaging batteries with fast charging?"

FEELS

- Financial pressure
- Frustrated by lack of centralized battery insights
- Concerned about operational disruptions
- Focused on optimization

SAYS

- "Battery cost is eating into margins."
- "We need predictive maintenance."
- "I can't afford unexpected breakdowns."

DOES

- Monitors fleet performance daily
- Uses fast charging to maintain schedules
- Reacts to issues after breakdowns



SENIOR BATTERY SYSTEMS ENGINEER

Name: Shweta Kulkarni

Age: 45

Role: Senior Battery Systems Engineer

Shweta oversees battery performance and reliability for an EV manufacturer. He works with R&D and warranty teams to improve battery lifespan and safety. However, limited real-world usage data makes it difficult to analyze degradation patterns and reduce warranty claims.

THINKS

- "Why are some batteries degrading faster?"
- "We need real-world usage data."
- "How can we reduce warranty costs?"

FEELS

- Accountable for battery reliability
- Frustrated by incomplete data
- Pressured to reduce warranty expenses
- Interested in advanced analytics

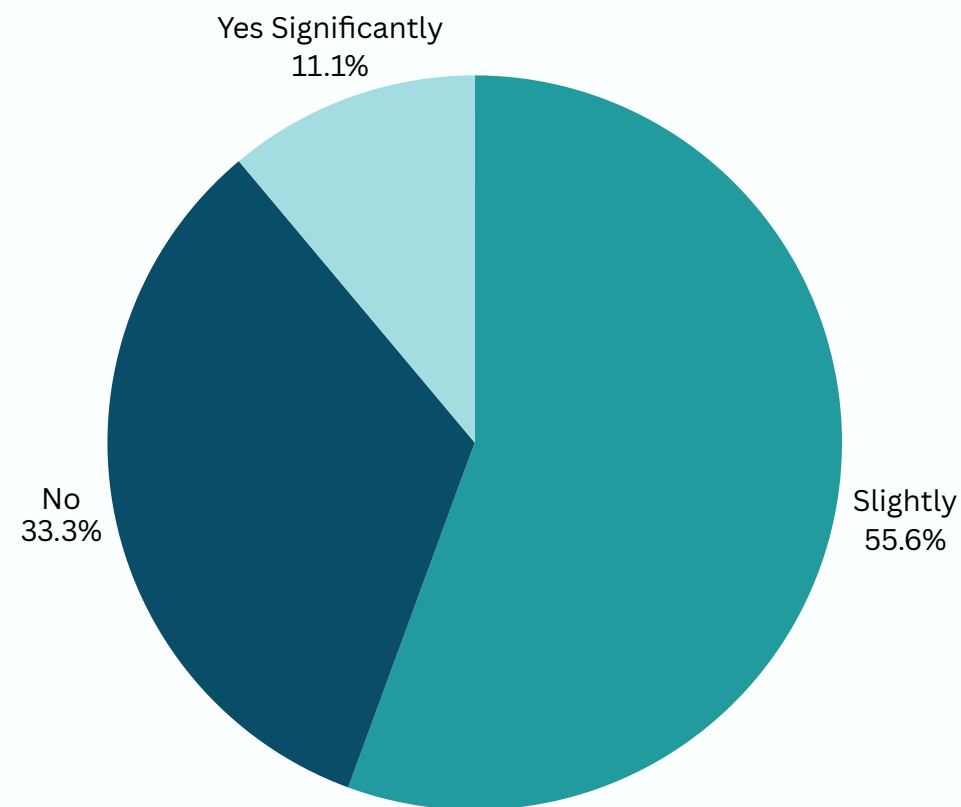
SAYS

- "We need better post-sale analytics."
- "We need accurate RUL prediction."
- "Data is fragmented."

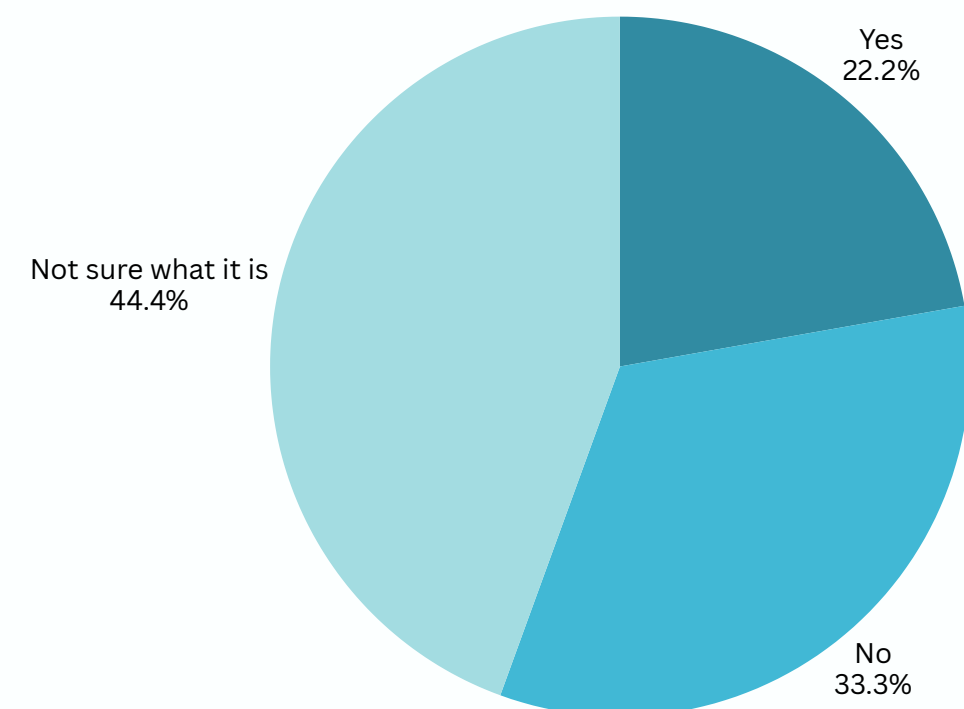
DOES

- Analyzes lab test results
- Reviews warranty claim reports
- Runs degradation simulations
- Relies on limited telemetry data

Have you noticed battery performance reducing over time?



Do you know your battery's State of Health (SOH)?



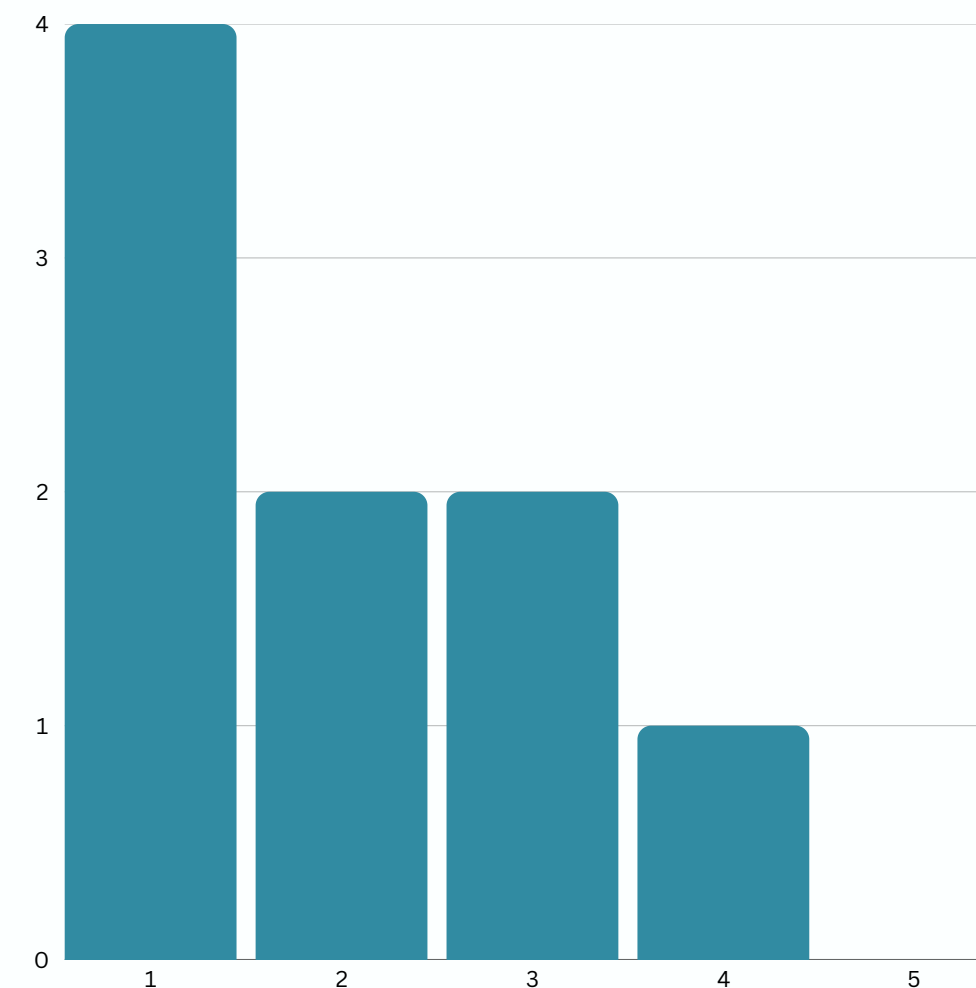
CORE SURVEY FINDINGS

Widespread Performance Decline: 66.7% of users have observed battery degradation over time, yet awareness of the actual State of Health (SOH) is critically low.

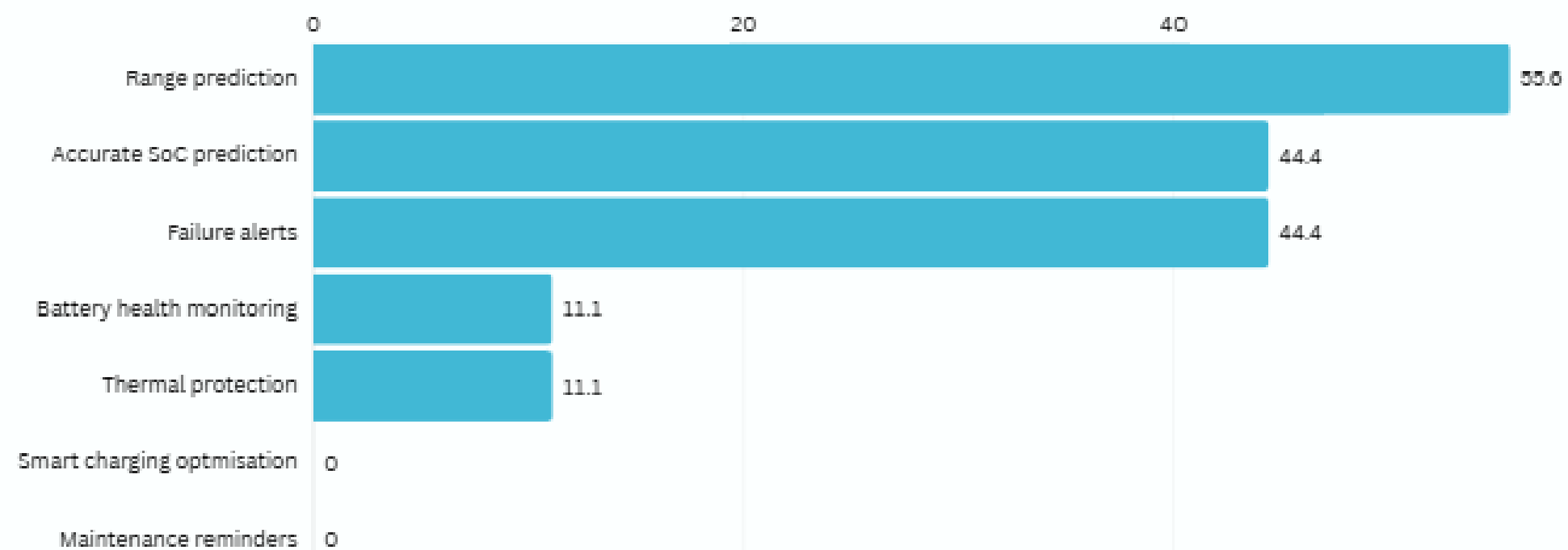
The "SOH" Knowledge Gap: Only 22.2% of respondents know their battery's SOH, while nearly half (44.4%) don't understand what the metric represents.

Trust Deficit in Data: Users frequently experience inaccurate battery percentage displays, indicating a need for more reliable monitoring systems.

How often does your displayed battery % feel inaccurate?



Which feature would you value most?



5

Market Analysis



EV Battery software market is projected to grow from \$6.3B–\$9.5B in 2025 to \$37B+ by 2035

Significant Growth: The EV battery market is expected to grow from USD 91.93 billion in 2024 to USD 251.33 billion by 2035.



The, shift towards wireless BMS (wBMS) and cloud-based analytics is a major trend, reducing weight and improving real-time monitoring

MARKET ANALYSIS

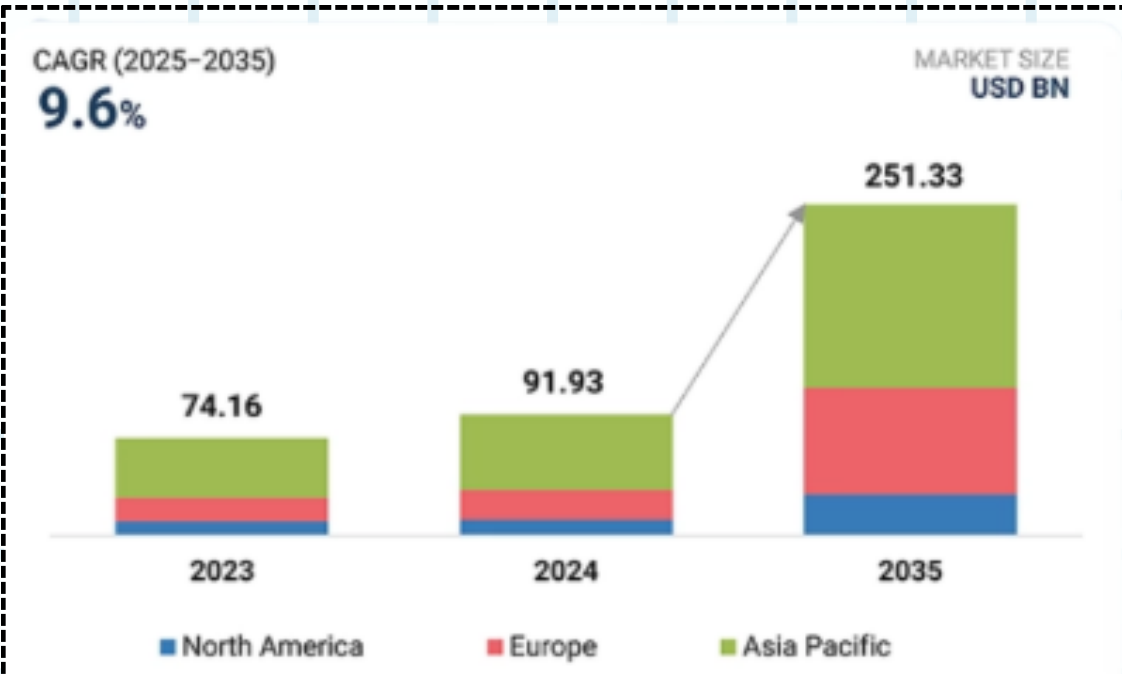
Growth Rate: This expansion represents a Compound Annual Growth Rate (CAGR) of 9.6%.



The BMS market is rapidly expanding, with, MarketsandMarkets forecasting a CAGR of 19.1%.

Government initiatives such as EU proposal for all city buses to be zero emission vehicles, surges demand for EV vehicles.

EV BATTERY: MARKET SIZE & SHARE



MARKET SNAPSHOT	
Market Size in 2024 (Value)	USD 91.93 BN
Market Forecast in 2035 (Value)	USD 251.33 BN
CAGR	9.6%
Years Considered	2019–2035
Base Year	2023
Forecast Period	2024–2035
Units Considered	Value (USD MN), Volume (Thousand Units)
Fastest-growing Region	Europe

 **CAGR 9.6%**
(2024–2035)

 **USD 251.33**
MARKET SIZE, 2035

6 COMPETITOR ANALYSIS

Features



SOC & Range Accuracy



Limited



Limited



Limited

SOH & RUL Prediction



Limited



Predictive Failure Detection



Limited



Limited

Smart Charging Optimization



Thermal Management Intelligence



Limited



Limited



Limited



Fleet & OEM Analytics Dashboard



Charging Only



Limited

Battery Buy-Sell Platform



AI Chatbot Assistant

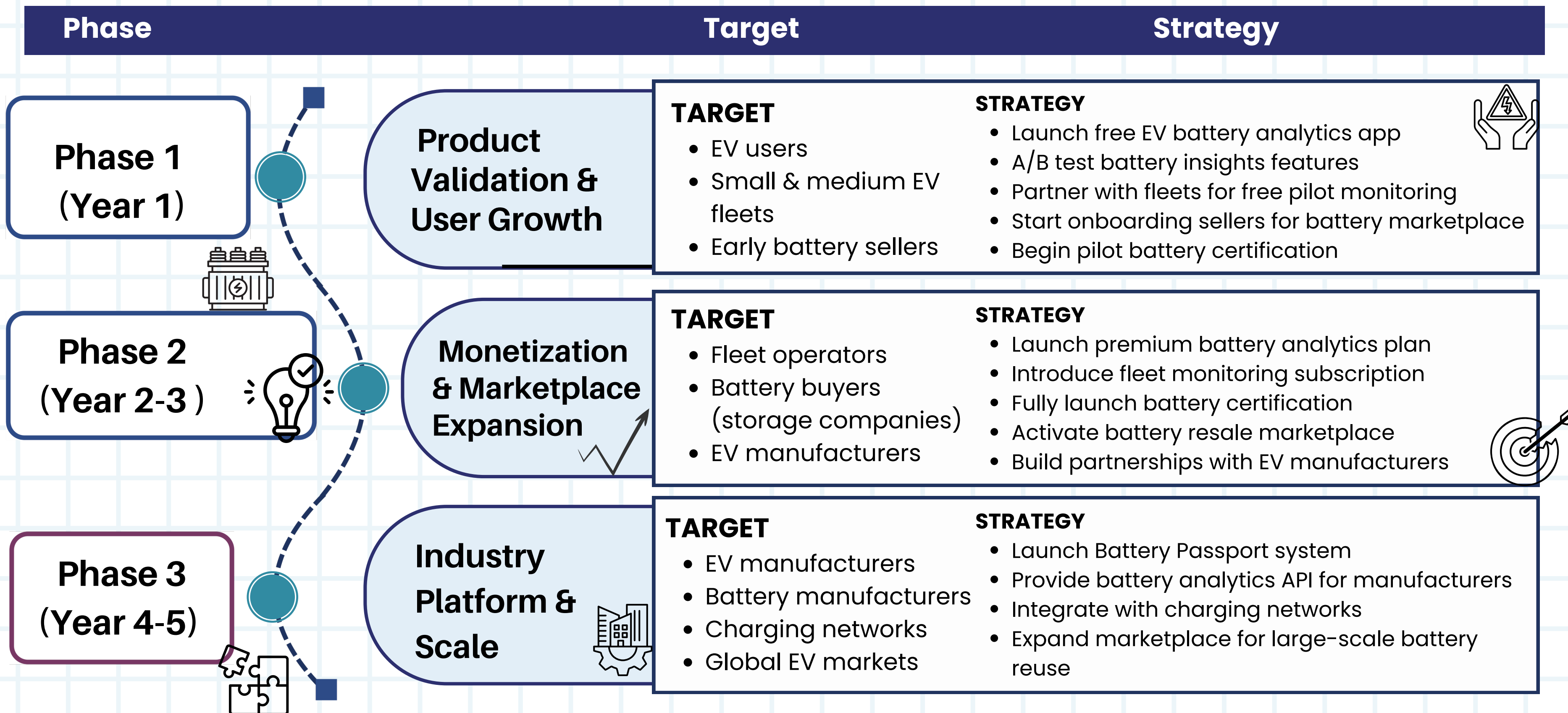


7

PRODUCT FEATURES



FEATURES	DESCRIPTION
BATTERY INTELLIGENCE AND INSIGHTS	This AI-driven platform leverages real-time telemetry —tracking voltage, temperature, and cell signals—to convert complex battery data into actionable insights like SOH, predictive range, and RUL . Through its Standard Plan (health scores and basic alerts) and Premium Plan (advanced degradation analytics and fleet-level predictive failure warnings), the system eliminates the "Trust Deficit" in the battery lifecycle . By providing a transparent " Battery Credit Score ," it effectively stabilizes resale value and optimizes asset longevity for both individual owners and fleet operators .
BATTERY OPTIMISATION SYSTEM	Uses temperature-aware, adaptive algorithms to optimize charging and extend battery life. By providing predictive maintenance and early failure detection , the system minimizes degradation, reduces wear, and prevents unexpected failures.
SECOND-LIFE BATTERY MARKETPLACE	Integrated platform enables transparent battery resale for second-life applications . Users can filter listings by health, capacity, and chemistry to match batteries with specific needs like stationary storage. This standardized marketplace reduces purchase risk and ensures condition-based pricing
BATTERY CERTIFICATION SYSTEM	Provides an independent verification layer that uses AI-based diagnostics and BMS data to ensure trust in battery resale . It generates tamper-resistant performance histories and digital health certificates to enable verified second-life deployment for buyers, fleets, and manufacturer



“ More batteries → better analytics → higher resale value → stronger marketplace liquidity.”



Cost Structure

- AI research & development
- Cloud infrastructure costs
- Data storage & processing
- Engineering team salaries
- Marketing & sales costs
- Platform maintenance



Customer Segments

- EV fleet operators
- Individual EV owners
- EV manufacturers (OEMs)
- Charging network operators
- Battery resale marketplaces



Customer Relationships

- SaaS subscription support
- Data analytics dashboard support
- Technical integration support
- AI chatbot assistance
- Customer success teams



Key Activities

- AI model development
- Battery data analysis
- Predictive analytics development
- Software platform development
- Charging optimization algorithms
- Data integration and telemetry
- Platform maintenance



Key Partners

- EV manufacturers (OEMs)
- Battery manufacturers
- EV charging networks
- Fleet operators
- Cloud service providers
- IoT/BMS hardware suppliers
- EV marketplaces / resale platforms



Value Propositions

- Accurate battery health insights
- Longer battery lifespan
- Reduced battery replacement cost
- Predictive failure prevention
- Smart charging recommendations
- Real-time battery intelligence
- Battery resale value transparency



Revenue Streams

- SaaS subscriptions (fleet operators)
- API licensing to OEMs
- Charging optimization software licensing
- Commission from battery buy-sell marketplace
- Enterprise analytics subscriptions
- Premium consumer app features



Key Resources

- AI battery analytics models
- Battery degradation datasets
- Data processing infrastructure
- Software platform & dashboards
- Battery modeling expertise
- Machine learning engineers



Channels

- Mobile application
- Web dashboard
- API integrations with BMS
- Fleet management software integration
- Partnerships with EV companies



REVENUE STREAM	PRICING MODEL
AI Battery Insights (Individual Users)	Premium analytics subscription – ₹99/month
Fleet Battery Analytics (SaaS)	Battery monitoring subscription – ₹199 per vehicle/month
Second-Life Battery Marketplace	Transaction commission on battery sales – ~10% per transaction
Bulk Battery Marketplace (B2B)	Commission on bulk battery deals – ~5–8% depending on volume
Battery Health Certification	Basic certificate – ₹399 Advanced AI diagnostic report – ₹799
Enterprise Data & API Licensing	Battery analytics data licensing for EV manufacturers, battery companies, and charging networks
Marketplace Services	Logistics coordination and platform service fees

11

SUSTAINABILITY AND IMPACT

SUSTAINABILTY

Pillar	Current pain point	Data-Driven Shift	Sustainable result
Value Loop	30% Margin Loss on manual testing and verification.	Battery credit score eliminates the diagnostic phase.	20–30% Cheaper BESS (Second-life) than new cells.
ERP Incentives	Regulatory Burden and heavy compensation fines	Compliance Bridge automates ERP certificate generation.	Tradable Financial Asset: Data itself funds the infrastrcuture
Urban Mine	Resource Blindness: Minerals lost to landfills.	Traceability ensures high-quality feedstock reaches recyclers.	Domestic Reserves: More efficient than mining lithium triangle.
Consumer TCO	70% Depreciation in 3 years due to trust deficit.	Verified Health Certs enable battery flipping and BaaS	0% green premium: Lower inerest rates and higher resale value.

Transaction Friction: By solving the Trust Deficit, we aren't just recycling batteries; we are creating a Marketplace for Energy Storage.

Self-Funding Loop: The ₹1,500 Cr incentive scheme shifts from a government cost to a private sector revenue stream once the data is verified.

IMPACT

Metric	Business-as-usual	With our AI platform
Resale value	30-40% residual value	50-65% residual value
Mineral import Dependency	High (Primary mining)	Reduced (via domestic recovery)
ERP Compliance	Fragmented and unverified	Automated and traceable
Safety Risk	Reactive (Post-failure)	Proactive (AI driven alerts)
BESS Adoption	Expensive (high testing cost)	Affordable (Data ready units))

12

Technical Aspects

BATTERY DATA ACQUISITION LAYER

PURPOSE:

Collect real-time battery data from vehicles and charging systems.

DATA SOURCES:

Battery Management System (BMS), vehicle sensors, charging stations, environmental sensors.

PARAMETERS COLLECTED:

Voltage, current, temperature, cell voltages, charging cycles, driving behavior, charging patterns.

CLOUD BATTERY INTELLIGENCE PLATFORM

Centralized data processing and analytics system.

FUNCTIONS

Real-time battery analytics, fleet battery monitoring, predictive maintenance, battery degradation tracking, manufacturer analytics insights.

OUTPUTS

Dashboards for EV users, fleet operators, battery manufacturers.

USER INTERFACES & APIS

MOBILE APP

Battery health score, range prediction, charging recommendations, battery usage insights.

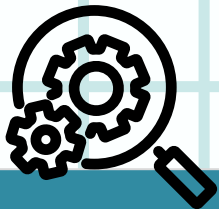
FLEET DASHBOARD

Battery health ranking, failure alerts, maintenance planning, fleet performance analytics.

INDUSTRY APIS

SOC prediction API, charging optimization API, battery analytics API for EV manufacturers, charging networks, battery companies.

BATTERY OPTIMIZATION ALGORITHMS



MODULE	TECHNICAL BASIS
Smart Charging Optimization	Dynamic charging current control based on battery condition and charge rate patterns
Thermal Management Prediction	Temperature monitoring and prediction to adjust charge/discharge limits
Cell Balancing Optimization	Detection of voltage imbalance across cells and optimization of balancing timing and current
Adaptive Battery Control	Adjustment of battery parameters using temperature data, driving behavior, and usage patterns

AI & MACHINE LEARNING ENGINE



MODEL	TECHNICAL BASIS
SOC Estimation Model	Uses voltage, current, and temperature telemetry from BMS with real-time charge/discharge data
SOH Prediction Model	Analyzes charging cycles, depth of discharge, internal resistance, and capacity fade trends
Range Prediction Model	Combines battery charge level, energy consumption, driving behavior, terrain, and weather data
Remaining Useful Life (RUL) Model	Uses cycle count, charge-discharge history, and battery aging patterns
Failure Prediction Model	Monitors voltage deviations, temperature spikes, and cell imbalance signals

13 Key Performance Indicator

Platform Adoption

Total EV users, monthly active users (MAU), vehicles connected to platform, fleet operators onboarded

Growth of the user and vehicle network

Battery Performance Impact

Reduction in battery failures, predictive maintenance alerts generated, improvement in battery lifespan, fleet maintenance cost savings

Real-world impact of the platform

Certification Adoption

Batteries certified, certification requests, certified batteries sold through marketplace, certification reliability score

Trust and adoption of the battery certification system

Battery Data Scale

Batteries monitored, battery telemetry data volume, battery health reports generated, AI prediction accuracy (SOC, SOH, RUL)

Strength of the data network and AI models

Revenue & Monetization

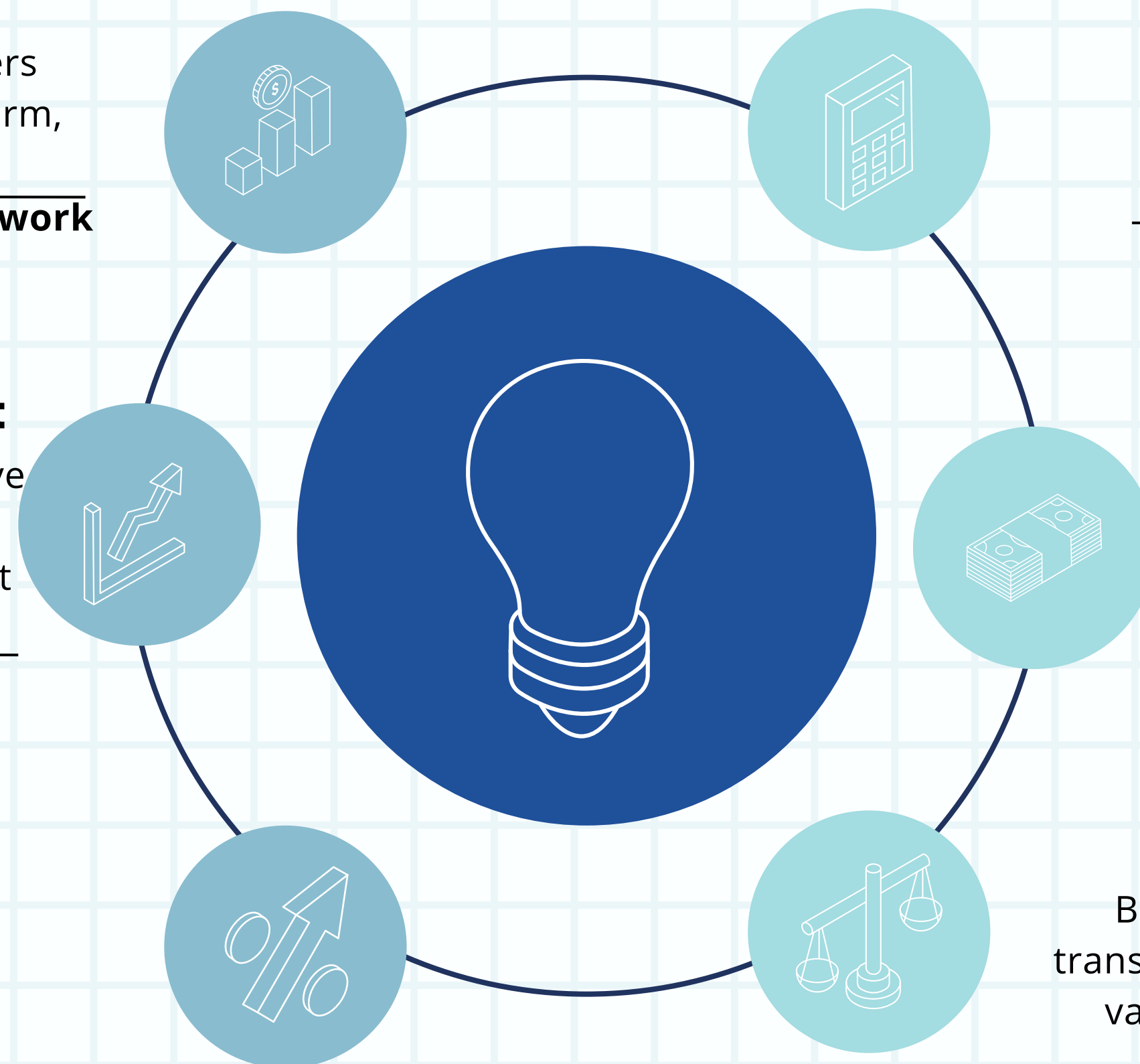
Premium subscriptions, certification revenue, marketplace transaction fees, enterprise/fleet analytics contracts

Financial performance and monetization

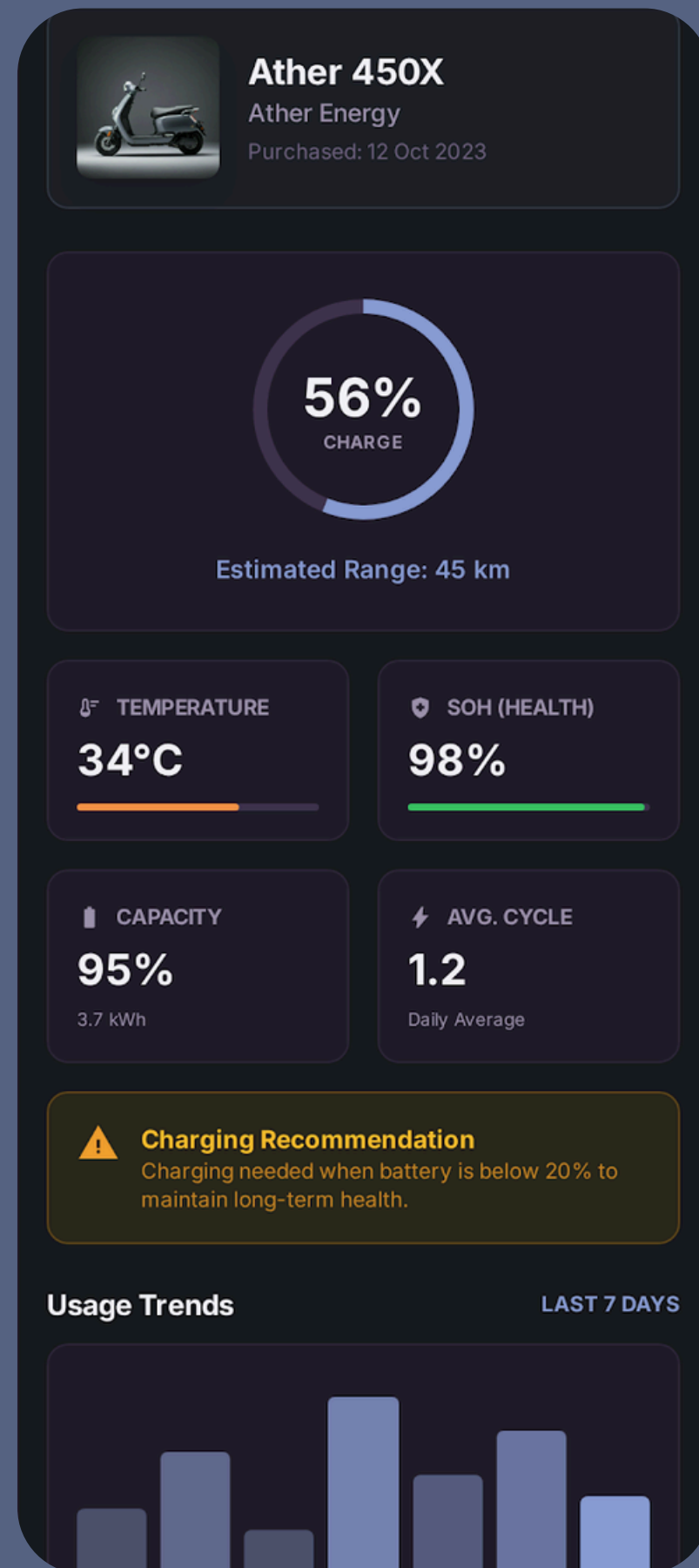
Marketplace Activity

Batteries listed for resale, completed transactions, marketplace GMV (transaction value), average resale price by battery health

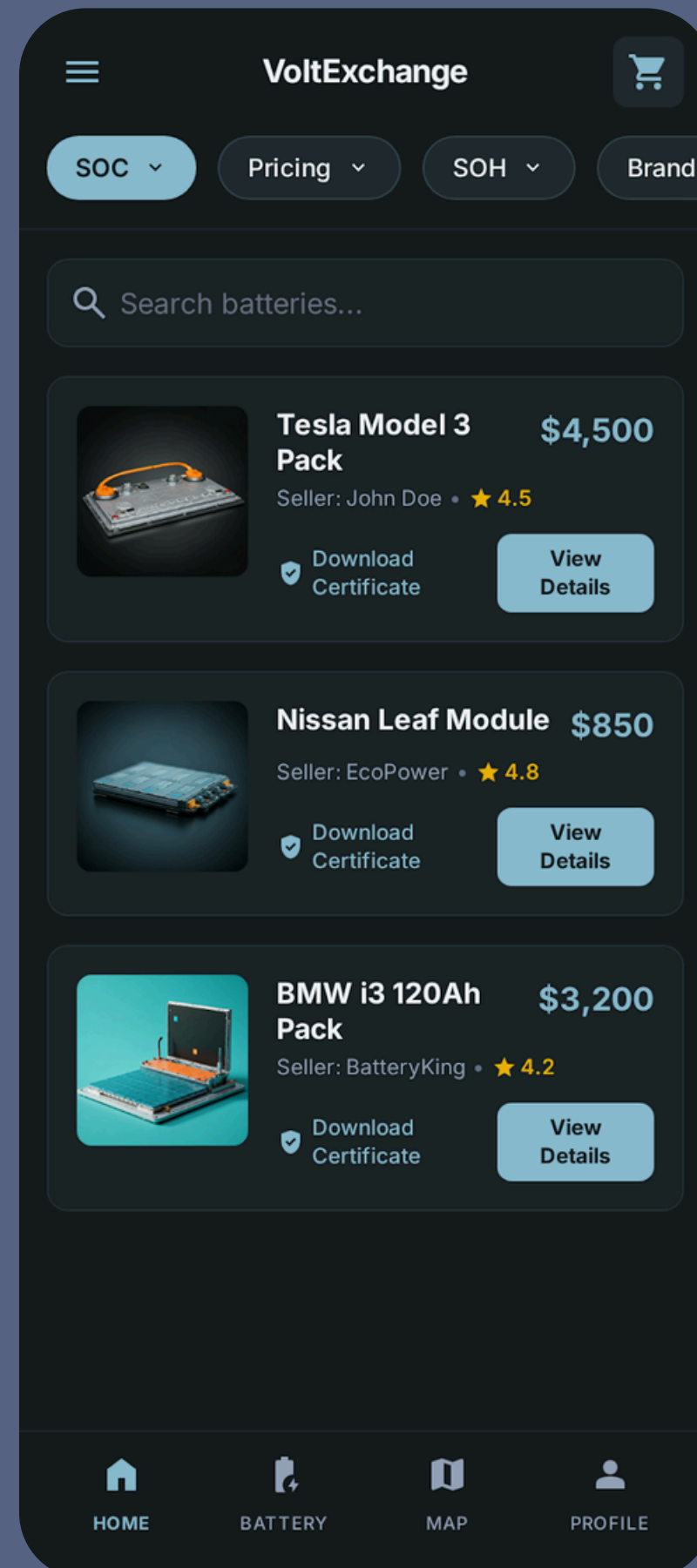
Liquidity and activity in second-life battery market



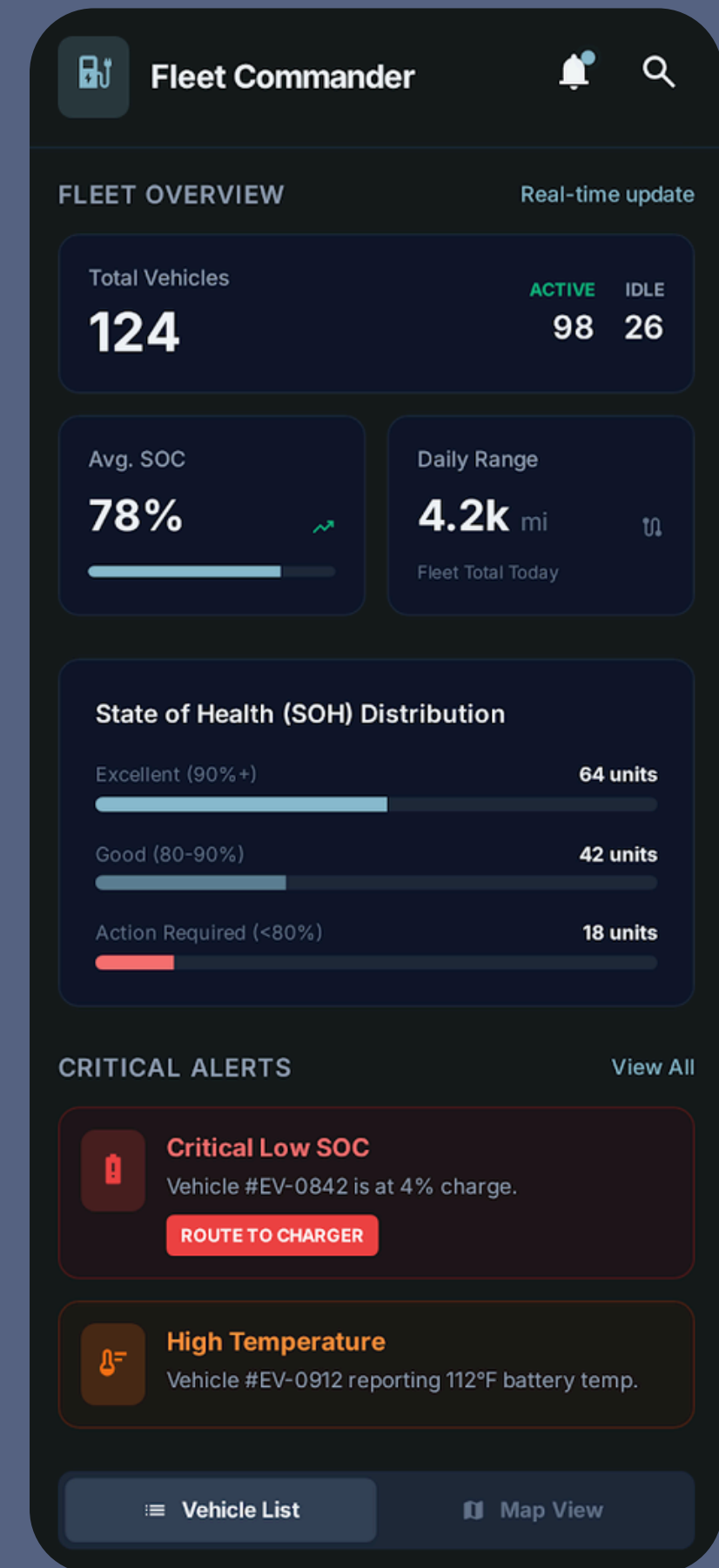
EV Battery Health Dashboard



Second-Life Battery Marketplace



Fleet Battery Monitoring Dashboard



THANK YOU